Smart phone gyaan
that no one gives you

We look under the hood of a smartphone and tell you everything you need to know about it's innards

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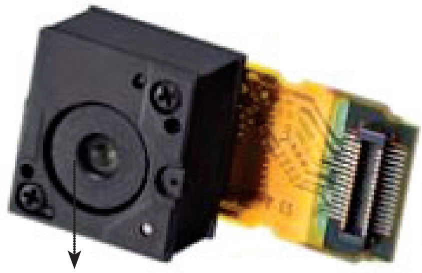
Screen
There are two factors when talking about displays. One is the kind of display technology used and the other is the kind of touch technology overlaying it. Although there is great hoopla about AMOLED displays in the market right now, a large section of phones still utilise LCD (Liquid Crystal Display) technology. Within the LCD space the most commonly found are TFT (thin-film-transistor) displays. TFTs don't have great viewing angles, have poor visibility in direct sunlight and consume relatively more power. The other type is IPS (in-plane switching) technology found in a few rare phones such as iPhone 4 and Droid X. Apple calls it 'retina' technology, which is merely marketing gimmickry. However IPS displays look brilliant (viewing angles et al) thanks to their crystal molecules being oriented such that their motion is parallel to the panel rather than perpendicular. AMOLED (Active-matrix organic light-emitting diode) displays are all the hype lately thanks to Samsung and HTC plonking them in phones such as Omnia II and Legend. In these displays organic compounds form the electroluminescent material making them quite vibrant and with great contrast. Super-AMOLEDs (found in Samsung Wave and Galaxy S) take things a step further to improve clarity by placing the touch sensors on the display itself rather than a separate layer. This makes it wafer thin and improves viewing angles. Speaking of touchscreens there are two types – resistive and capacitive. Resistive, the more primitive technology, works by placing two layers of conductive material close to each other. When a finger or stylus depresses the layer on top, it makes a contact with the other layer forming a circuit at that location. Capacitive is the more advanced technology and is now increasingly being used. For these screens the glass (an insulator) is coated with a transparent conductor such as indium tin oxide. When a finger touches the screen it breaks the electrostatic field. It's position is then measured relative to the four corners of the screen by the processor.



Camera
Smart phone cameras depend a lot on image processing rather than just the hardware when it comes to image quality. This is because no matter how large the camera resolution gets in Megapixels (MP), but the sensor size remains small you will have bad imaging. The sensor is what converts the optical image into electrical signal. What matters also is the autofocusing capability, shutter speed control etc.

Microphone
Smartphone microphones depend on chipset circuitry and processing to clear out the vocal signal. It can be termed as noise suppression or cancellation depending on the technology used. Cutting edge Smartphones (iPhone 4 and Droid X for eg) use two microphones for cancelling out noise. The second one is placed far from the voice mic. It detects the signature of ambient noise and suppresses it.